

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – SUB-STRAND 3.2 PROBABILITY I: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Answer all five questions in the space under each prompt. Remember the puzzle we began with: during the 2023 Kenyan Premier League season, fans said Gor Mahia FC was 'due' for a loss after a winning streak, while betting advertisements on Nairobi matatus and social media told youth they could 'beat the odds'. Use what you learned in Lessons 1 to 8: your coin, die and token experiments, the probability formula, the range of probability, mutually exclusive events and independent events. Show all working, write probabilities as fractions in their simplest form, and explain your reasoning in full sentences. Where a question says 'suppose', use the numbers given for that question only.

Part 1: Does the past change the chance? (Equally likely events)

In our class experiment, each pair tossed a 10-shilling coin 40 times and we pooled the results. A fan at a Nairobi bus stage says: 'A coin has landed on heads five times in a row, so tails is now more likely.'

- (a) What is the theoretical probability of heads on one toss of a fair coin? Show how you get it using the formula.
- (b) Is the fan correct? Explain your answer using what you know about how one toss relates to the next.
- (c) Why did pooling the whole class's results give a more reliable estimate of the probability than your own pair's 40 tosses?
- (d) Name the mistake in thinking that the fan, and the fans who said Gor Mahia was 'due' for a loss, are making.

- (a) $P(\text{heads}) = \text{number of favourable outcomes} \div \text{total number of equally likely outcomes} = 1 \div 2 = 1/2$ (0.5 or 50%).
- (b) No. Each toss is an independent event: the result of one toss gives no information about the next. A coin has no memory, so after five heads the probability of tails on the next toss is still $1/2$, exactly the same as the probability of heads.
- (c) A larger sample gives a more reliable estimate. One pair's 40 tosses can drift away from $1/2$ by chance, but when many pairs' results are pooled these chance differences tend to even out and the experimental probability gets closer to the theoretical $1/2$.
- (d) This is the gambler's fallacy: believing that past outcomes change the probability of future independent events. Each match Gor Mahia plays is treated as an independent event, so the earlier wins do not change the probability of the next result. The fans who say the team is 'due' for a loss are wrongly treating the next match as if it depends on the earlier wins.

Part 2: Computing probabilities from the mandazi bag (Equally likely outcomes and the range of probability)

A cloth bag at a school canteen holds 15 tokens that represent mandazi flavours: 6 plain (white), 5 coconut (yellow) and 4 cardamom (green). One token is drawn at random.

- (a) Calculate $P(\text{plain})$, $P(\text{coconut})$ and $P(\text{cardamom})$ as fractions in simplest form.
- (b) Add the three probabilities. What does your total tell you about the possible outcomes of one draw?
- (c) Write down one event for this bag that has probability 0, and one that has probability 1.
- (d) A betting advertisement claims '120% chance of winning'. Using the range of probability, explain whether this claim can be true.

- (a) $P(\text{plain}) = 6/15 = 2/5$. $P(\text{coconut}) = 5/15 = 1/3$. $P(\text{cardamom}) = 4/15$.
- (b) $6/15 + 5/15 + 4/15 = 15/15 = 1$. The three flavours cover every possible outcome of one draw, so one of them is certain to happen. The probabilities of all the individual outcomes add up to 1.
- (c) Probability 0 (impossible): drawing a red token, since the bag has no red tokens ($P = 0/15 = 0$). Probability 1 (certain): drawing a token that is not red, since every token in the bag is white, yellow or green ($P = 15/15 = 1$).
- (d) No. Every probability must satisfy $0 \leq P \leq 1$. 120% is the same as 1.2, which is greater than 1, so it is mathematically impossible. No winning streak or advertisement can push a probability outside this range.

Part 3: Mutually exclusive events and the addition rule

- (a) Use the mandazi bag from Part 2. Are the events 'draw a plain mandazi' and 'draw a coconut mandazi' mutually exclusive on a single draw? Explain what mutually exclusive means.
- (b) Calculate $P(\text{plain or coconut})$ for one draw and show your rule.
- (c) In our token experiment each group drew 20 times with replacement. Why might a group's experimental result for 'plain or coconut' be close to, but not exactly equal to, your answer in (b)?

- (a) Yes. Mutually exclusive events cannot both happen on a single trial. One draw gives only one token, so it cannot be both plain and coconut.
- (b) For mutually exclusive events, $P(A \text{ or } B) = P(A) + P(B)$. So $P(\text{plain or coconut}) = 6/15 + 5/15 = 11/15$.
- (c) The experimental probability is based on only 20 draws, and chance causes small differences from the theoretical value. With more draws the experimental result would get closer to 11/15.
- (d) Win and Draw cannot both happen in one match, so they are mutually exclusive and $P(\text{Win or Draw}) = 1/3 + 1/3 = 2/3$. Since the three results cover everything, $P(\text{Loss}) = 1 - 2/3 = 1/3$. An advertisement that highlights 2/3 hides this remaining probability of a Loss, the outcome where the bettor loses. The addition rule must not be used when events can happen together (overlap), because adding would count the overlap twice and give an inflated probability. It also only gives correct answers when each individual probability has been correctly found.

(d) Suppose that, for one Gor Mahia match, we assume $P(\text{Win}) = 1/3$ and $P(\text{Draw}) = 1/3$ and that Win, Draw and Loss are the only possible results. (Real matches need not have equal chances; use these numbers for this question only.) Calculate $P(\text{Win or Draw})$. Then explain why a betting advertisement that only talks about 'Win or Draw' could mislead a bettor, and state when the addition rule must not be used.

Part 4: Independent events, the multiplication rule and why the stall always wins

(a) Suppose that for each match the probability that Gor Mahia wins is $1/3$ (an assumed value for this question), and each match is independent. Calculate the probability that Gor Mahia wins two matches in a row, and then three matches in a row. Show the rule you use.

(b) Gor Mahia has just won five matches in a row. A fan says the probability of winning the sixth match must now be different from $1/3$. Use the idea of independent events to respond.

(c) A betting stall in Nairobi charges Ksh 20 per spin. A player wins Ksh 30 whenever the spinner lands on the winning sector, and $P(\text{win}) = 1/3$ on each spin. Over 180 spins, a win is expected on about $1/3$ of the spins. Using this, work out the money you would expect the stall to collect, the money you would expect it to pay out, and the profit you would

(a) For independent events, $P(A \text{ and } B) = P(A) \times P(B)$. Two wins in a row: $1/3 \times 1/3 = 1/9$. Three wins in a row: $1/3 \times 1/3 \times 1/3 = 1/27$. The probability of a longer streak gets smaller with each extra match.

(b) The fan is wrong. Each match is an independent event, just like each coin toss, so the five-match streak carries no information about match six. The probability of winning the sixth match is still the same fixed $1/3$. The streak does not raise it or lower it.

(c) Collected: $180 \times \text{Ksh } 20 = \text{Ksh } 3,600$. Expected number of wins: $1/3 \times 180 = 60$. Expected payout: $60 \times \text{Ksh } 30 = \text{Ksh } 1,800$. Expected profit for the stall: $\text{Ksh } 3,600 - \text{Ksh } 1,800 = \text{Ksh } 1,800$. This is what we expect on average over many spins, not a guarantee for any short run: a player can have a lucky streak, but over many spins the stall is expected to collect more than it pays out. The player wins only $1/3$ of the time yet pays Ksh 20 every spin, so the game is set up to disadvantage the player, and gambling is a losing proposition.

expect. Explain what this shows about gambling.	
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Part 5: Answering the Driving Question

Driving question: If past outcomes of a chance event, like a coin toss or a football match, do not change future probabilities, how can we use mathematics to predict what is likely to happen, and why does this understanding protect us from the dangers of gambling? Write a full explanation (about 8 to 10 sentences) that: – uses all four ideas from this unit: equally likely events, the range of probability, mutually exclusive events and independent events; – refers to the Gor Mahia winning-streak debate and the betting advertisements on Nairobi matatus and social media; – ends by saying how this knowledge can help you make a responsible decision.	Mathematics lets us say what is likely, not what will certainly happen. Independence does not stop us doing this: each trial has a fixed probability, and we can calculate it and compare it with results over many trials. For equally likely outcomes we use $P(\text{event}) = \text{favourable outcomes} \div \text{total outcomes}$, for example $P(\text{heads}) = 1/2$ for a fair coin, and our pooled class data supported this because larger samples give more reliable estimates. Every probability lies between 0 and 1, so an advertisement claiming '120% odds' or a commentator saying a streak makes a win 'inevitable' is mathematically impossible. When outcomes are mutually exclusive, such as Win, Draw and Loss in one Gor Mahia match, we add their probabilities, but advertisements that stress only the favourable combination, like $P(\text{Win or Draw})$, hide the probability of losing. Most importantly, each match is an independent event, so a coin or a team has no memory: the probability of the next match stays fixed no matter how long the winning streak is. The fans who said Gor Mahia was 'due' for a loss, and bettors who think tails is 'due' after five heads, are making the gambler's fallacy. The multiplication rule shows that the probability of winning several bets in a row gets smaller with each extra bet, so long winning streaks are unlikely. In our stall example, the stall expected to collect Ksh 3,600 and pay out Ksh 1,800 over 180 spins, so it expects a profit, although short runs can differ. Understanding this protects me because I will not trust betting advertisements that promise I can 'beat the odds', and I will not spend my money chasing a 'due' win. Instead I will use probability to make careful and responsible decisions.
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RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Correct use of probability calculations (formula, addition rule, multiplication rule)	All calculations are correct and in simplest form, the correct rule is named and used each time, and working is clearly shown.	Most calculations are correct with the correct rule used; there may be one small arithmetic slip or missing step of working.	Several calculations are incorrect, or rules are mixed up (for example adding when events should be multiplied), or working is missing.
Understanding of independence and the gambler's fallacy	Clearly explains that each independent event has a fixed probability that past results cannot change, names the	States that past outcomes do not change future probability and applies it to the coin or the football example, but the	Does not reject the idea that a team or coin is 'due', or gives an explanation that is unclear or incorrect.

	gambler's fallacy, and applies it to both the coin and Gor Mahia.	explanation is less complete.	
Understanding of the range of probability and mutually exclusive events	Correctly explains that $0 \leq P \leq 1$, that probabilities of all outcomes sum to 1, and when events are mutually exclusive and when the addition rule can and cannot be used.	Shows the range and the addition rule correctly but gives limited explanation of when the rule does not apply.	Confuses the range of probability or the meaning of mutually exclusive, or cannot say when the addition rule applies.
Connection to the Driving Question and the Kenyan context	Combines all four concepts in a coherent argument, explains how independence is compatible with predicting what is likely, uses the Gor Mahia debate, betting advertisements and the stall data accurately, and ends with a thoughtful responsible decision.	Uses most concepts and the Kenyan context correctly, and gives a reasonable conclusion about responsible decisions.	Gives a general answer with few concepts, little or no reference to the Kenyan phenomenon, or no conclusion about decision-making.
Communication and reasoning	Writing is clear, logical and uses correct mathematical vocabulary in full sentences throughout.	Writing is mostly clear with mostly correct vocabulary.	Writing is hard to follow or key vocabulary is missing or misused.

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